Unit: Unit 0: Foundations of Algebra
Topic: Inequalities

Name: ________________
Date: ___________

Homework (Carolina Reaper)

Q1. Solve for x in $2x - 5 > 11$.
(A) $x > 8$
(B) $x < 8$
(C) $x > 4$
(D) $x < 4$
(E) $x = 8$

Q2. Graph the solution to $x + 2geq7$ on a number line.
(A) $[5, infty)$
(B) $(-infty, 5]$
(C) $[7, infty)$
(D) $(-infty, 7]$
(E) $[2, 7]$

Q3. A music producer needs to mix a track with a frequency range of $200leqfleq800$ Hz. Write the solution in interval notation.
(A) $[200, 800]$
(B) $(200, 800)$
(C) $[200, 800)$
(D) $(200, 800]$
(E) $(-infty, 200]cup[800, infty)$

Q4. An artist is creating a canvas with a width w that satisfies $w - 3 > 10$. Solve for w.
(A) $w > 13$
(B) $w < 13$
(C) $w > 7$
(D) $w < 7$
(E) $w = 13$

Q5. A rhythm pattern has a tempo t that must satisfy $t + 2leq120$. Write the solution in interval notation.
(A) $(-infty, 118]$
(B) $[118, infty)$
(C) $[0, 118]$
(D) $(118, infty)$
(E) $(-infty, 118)$

Q6. Solve for x in $x/4 - 2 < 5$.
(A) $x < 28$
(B) $x > 28$
(C) $x < 14$
(D) $x > 14$
(E) $x = 28$

Q7. A DJ needs to adjust the bass level b to satisfy $b + 5geq20$. Write the solution in interval notation.
(A) $[15, infty)$
(B) $(-infty, 15]$
(C) $[15, 20]$
(D) $[20, infty)$
(E) $(-infty, 20]$

Q8. Graph the solution to $3x - 2leq14$ on a number line.
(A) $(-infty, 16/3]$
(B) $[16/3, infty)$
(C) $[14/3, infty)$
(D) $(-infty, 14/3]$
(E) $[2, 14/3]$

Open Ended Questions (FRQ)

Q9. A painter is creating a mural with a height h that satisfies $h - 4 > 2h - 10$. Solve for h and write the solution in interval notation. Show all work.

Q10. A sound engineer needs to adjust the treble level t to satisfy $2t + 5geq11$. Write the solution in interval notation and graph the solution on a number line. Show all work.

Chef's Tips

- Tip 1: Ensure students use the correct inequality signs when solving and graphing.
- Tip 2: Encourage students to check their work by plugging in values from the solution set.
- Tip 3: Remind students to label their graphs with the correct interval notation.